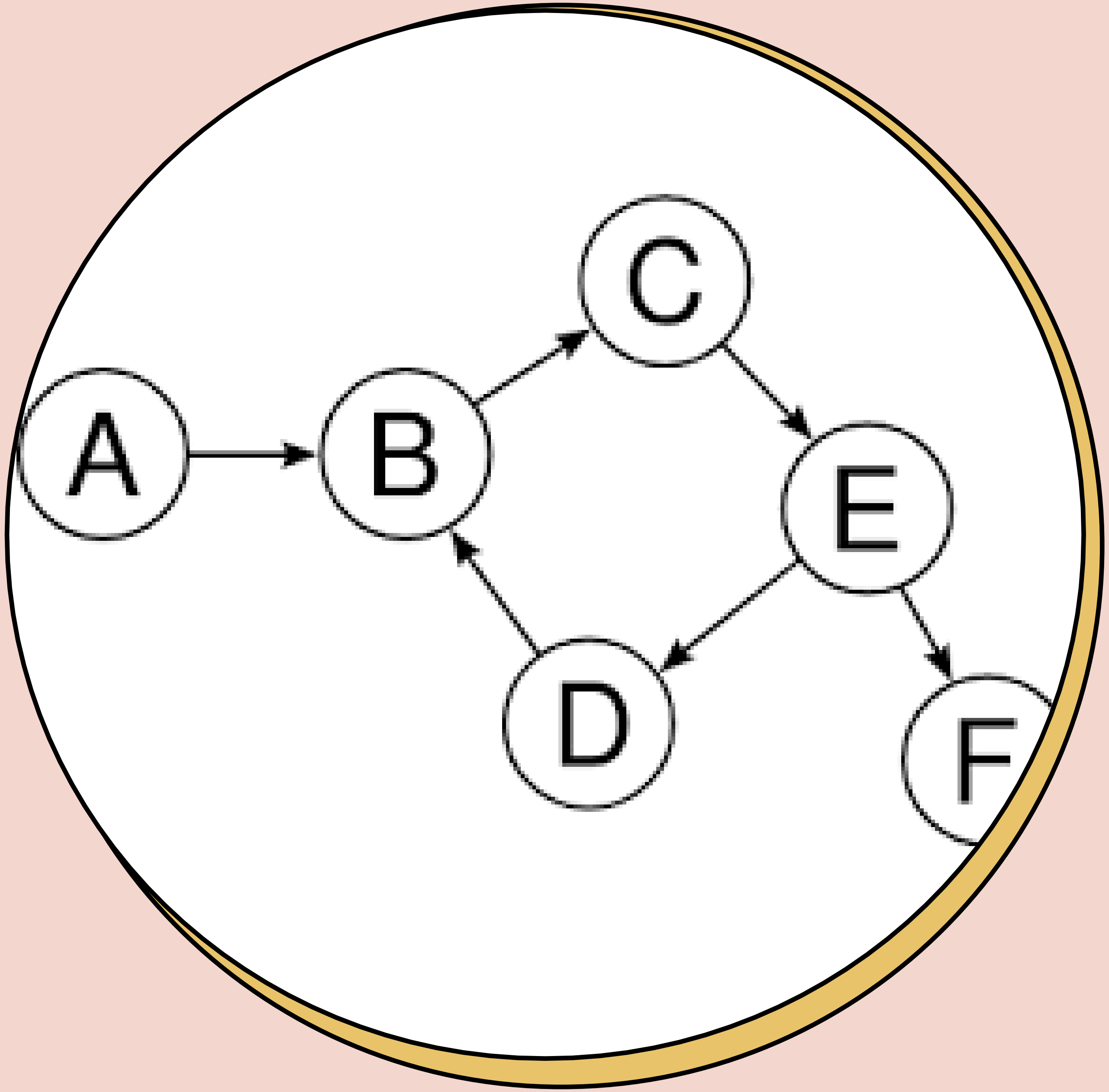
CASH FLOW MINIMIZER

A **cash flow minimizer** is an algorithm used to reduce the number of transactions required to settle debts or balances. The idea is to minimize the complexity and number of payments between individuals in a group, while ensuring that everyone's balance is settled.

DATASTRUCTRE USED

GRAPH



INPUTS

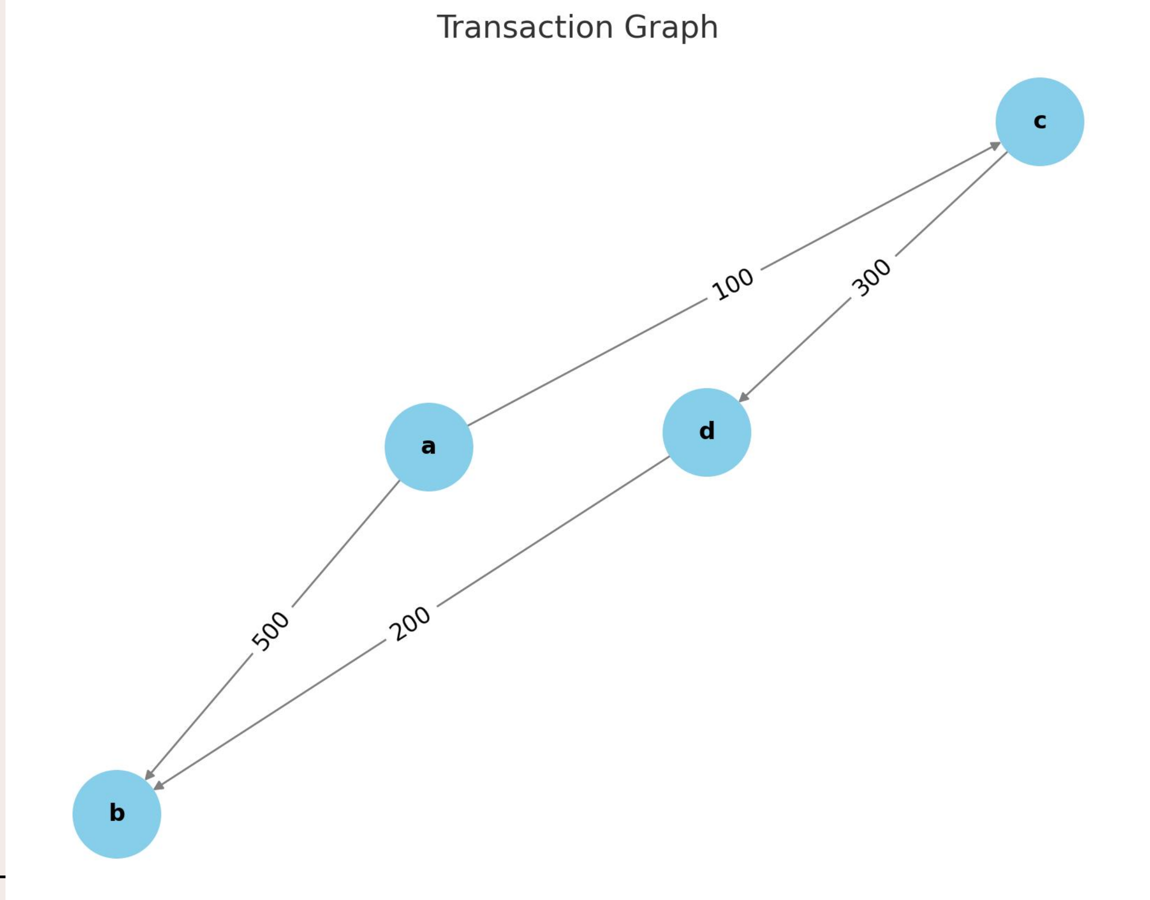
- Number of people
- Names number of payment options: payment options (First person has all modes of transactions).
- Total number of transactions:
- Details of transactions: debtor creditor amount

OUTPUT

- Min number of transactions
- Transactions: _ person to _ AMOUNT

CODE WORKING

- An adjacency matrix is created from the input



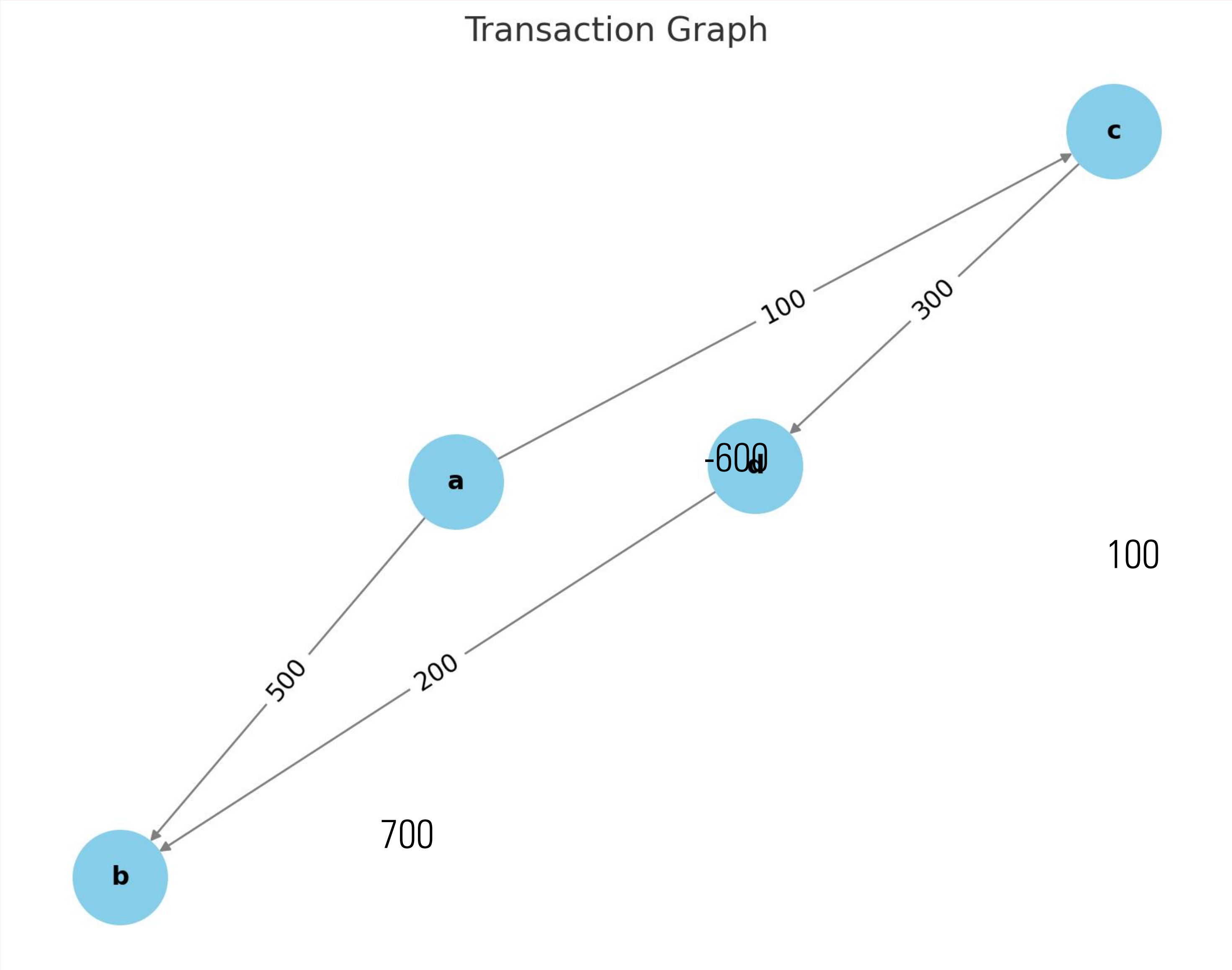
	a	b	c	D
A	0	500	100	0
B	0	0	0	0
C	0	0	0	300
D	0	200	0	0

For min number of transactions

- One transaction -> atleast one person is settled
- If no common mode of payment we call the leader- 2 transactions occur

working

- We check net cashflow per person
- If net cashflow= 0. $ is settled
- -ve owes money, +ve receives money
- Take the most negative amount, and find max value with common payment mode; else take max value
- Consider a transaction settling one of the participant
- Appropriately adjust the values.
- REPEAT



-200

-600

100

700

	a	b	c	D
A	0	500	100	0
B	0	0	0	0
C	0	0	0	300
D	0	200	0	0

A	B	C	D
-600	700	-200	100

Transaction between A and B: A->B 600

A	B	C	D
-600	700	-200	100
+600	-600	0	0

A	B	C	D
0	100	-200	100

A	B	C	D
0	100	-200	100

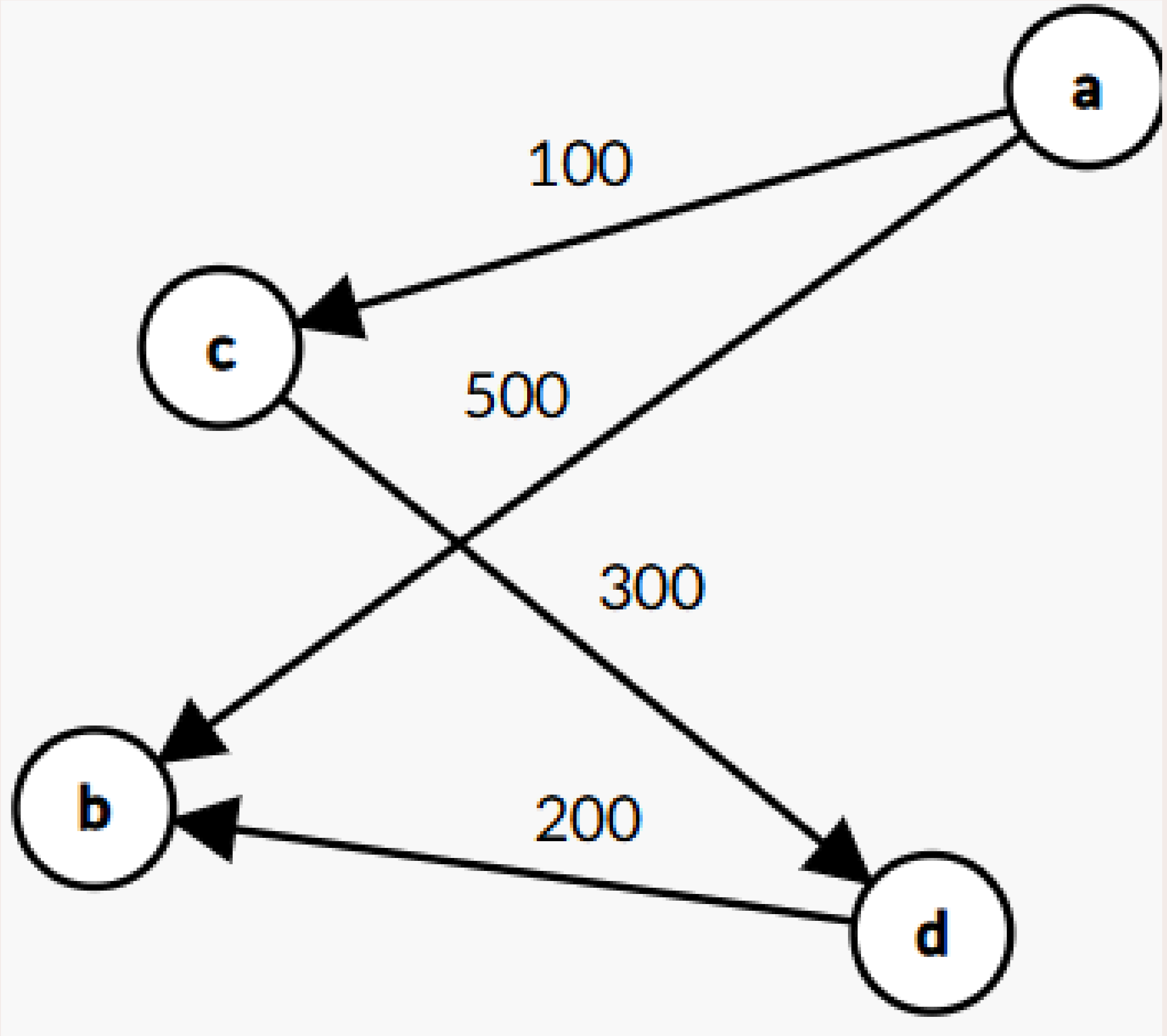
Min is C and max is B. C->B 100

A	B	C	D
0	0	-100	100

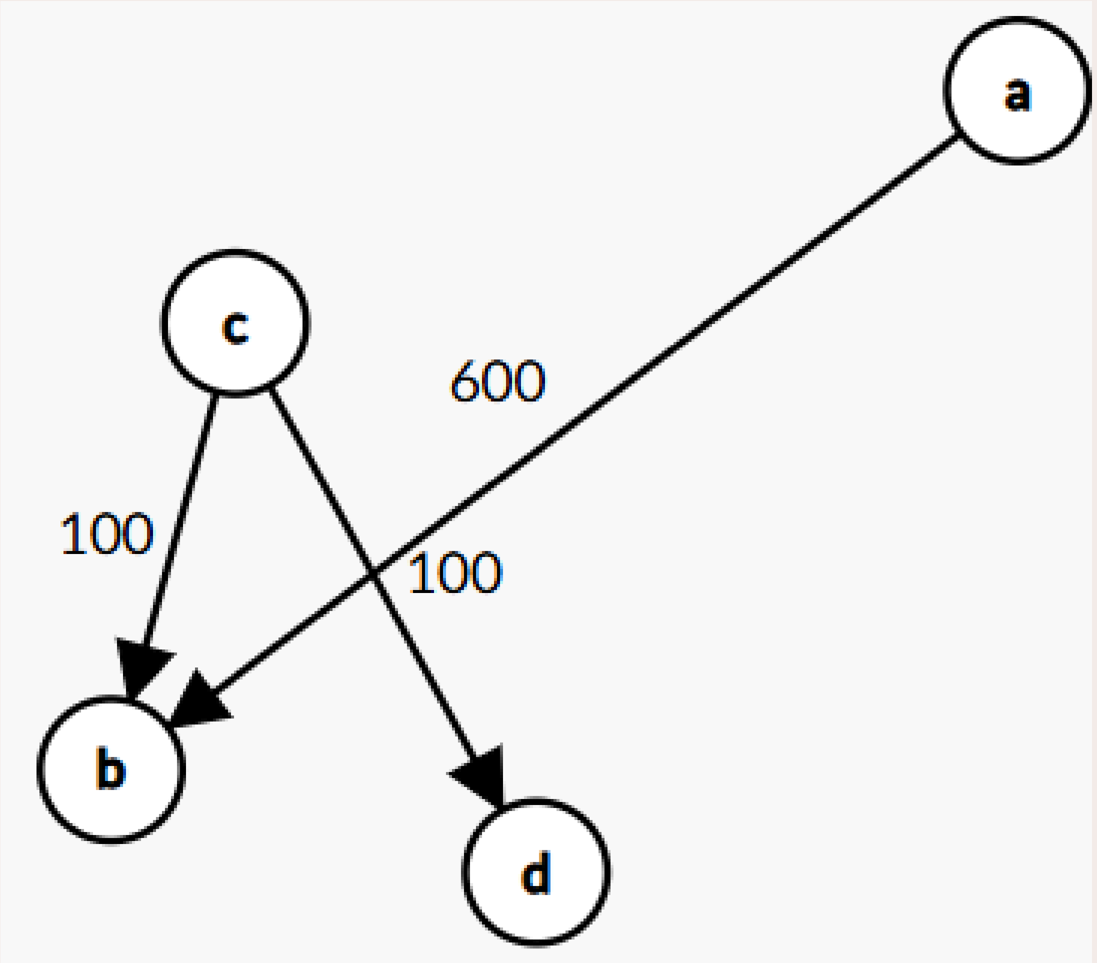
Min is C and max is D. C->D 100

A	B	C	D
0	0	0	0

- a b 500
- a c 100
- c d 300
- d b 200



- a b 600
- c b 100
- c d 100



2024

THANK YOU

DSA PROJECT